

Cross-Field Business Rule Engine — Knowledge Base (KB) Reference

Generated directly from `crossfieldruleengine.py ( build generic_kb() )`. **49 rules.**

By framework: IFRS9 = 11, IRB = 36, IRB+IFRS9 = 2

By rule type: Conditional = 13, Date ordering = 3, Domain/bound = 26, Identity/derivation = 3, Inequality = 4

By severity: CRITICAL = 6, MATERIAL = 21, MINOR = 22

Rules are written in **semantic roles**, not physical column names; the engine resolves roles to columns at run time. Each rule ships: an IF-condition (scope), an expression (compliant when True), a severity, optional encoded exceptions, and — where applicable — a regulatory reference. Verdict per rule: **PASS** (violation rate $\leq$ tolerance, default 0), **VIOLATION**, or **NOT-APPLICABLE** (empty scope or unmapped role).

IRB (36 rules)

ID	Sev	Type	Entity	Roles (semantic)	Rule	Reg reference
IRB-01	CRIT	Inequality	workout	exposureatdefault, drawnbalance	exposureatdefault >= drawnbalance (EAD cannot be below drawn balance)	—
IRB-02	CRIT	Date ordering	workout	possessiondate, saledate	possessiondate <= saledate	—
IRB-30	CRIT	Date ordering	workout	defaultdate, possessiondate	defaultdate <= possessiondate (default precedes repossession)	CRR Art. 178; EBA/GL/2016/07
IRB-31	CRIT	Date ordering	workout	defaultdate, saledate	defaultdate <= saledate (default precedes collateral sale)	EBA/GL/2017/16 (LGD, recovery timing)
IRB-03	MATE	Conditional	workout	workoutstate, realisedlgd	IF workout closed/writtenoff THEN realisedlgd populated (open=censored)	—
IRB-04	MATE	Conditional	workout	workoutstate, saledate, cureindicator	IF workout closed AND not cured THEN saledate populated (exceptions: cureindicator=1)_	—
IRB-05	MATE	Inequality	workout	realisedloss, exposureatdefault, workoutstate	realisedloss <= exposureat_default (cannot lose more than the exposure)	—
IRB-06	MATE	Identity/derivation	workout	realisedlgd, realisedloss, exposureatdefault, workoutstate	realisedlgd == realisedloss / exposureat_default (+/- 0.01)	—
IRB-07	MATE	Conditional	workout	cureindicator, workoutstate	IF cured THEN workoutstate != writtenoff (cure/write-off exclusive)	—
IRB-15	MATE	Domain/bound	workout	workoutstate	workoutstate in {closed, open, written_off}	—
IRB-16	MATE	Identity/derivation	facility	originalltv, originalbalance, originalpropertyvalue	originalltv == originalbalance / originalpropertyvalue * 100 (+/- 0.5)	—
IRB-17	MATE	Domain/bound	facility	originalltv	originalltv in [0, 200] (logical bound; negative LTV impossible)	—

ID	Sev	Type	Entity	Roles (semantic)	Rule	Reg reference
IRB-25	MATE	Domain/bound	borrower	ageatorigination	ageatorigination in [18, 100] (logical bound)	—
IRB-27	MATE	Domain/bound	valuation	valuationamount	valuationamount >= 0 (logical bound)	—
IRB-32	MATE	Conditional	workout	workoutstate, timetoresolution	IF workout closed THEN timeto_resolution present	EBA/GL/2017/16 para 158 (realised LGD)
IRB-33	MATE	Inequality	workout	exposureatdefault, originalbalance	exposureatdefault <= originalbalance * 1.5 (EAD bounded vs original)	CRR Art. 166 (exposure value)
IRB-34	MATE	Inequality	workout	drawnbalance, originalbalance	drawnbalance <= originalbalance * 1.5 (balance-at-default vs original)	CRR Art. 166; amortisation sanity
IRB-35	MATE	Domain/bound	workout	realisedlgd	realisedlgd in [-0.10, 1.50] (downturn LGD plausibility bound)	EBA/GL/2017/16 (LGD)
IRB-36	MATE	Conditional	workout	cureindicator, realisedlgd	IF cured THEN realised_lgd <= 0.05 (a genuine cure implies ~no loss)	EBA/GL/2017/16 (cure definition)
IFRS9-10	MINO	Domain/bound	snapshot	monthsonbook	monthsonbook >= 0 (logical bound)	—
IRB-08	MINO	Domain/bound	workout	legalcosts	legalcosts >= 0 (logical bound: cannot be negative)	—
IRB-09	MINO	Domain/bound	workout	othercosts	othercosts >= 0 (logical bound: cannot be negative)	—
IRB-10	MINO	Domain/bound	workout	saleproceeds	saleproceeds >= 0 (logical bound: cannot be negative)	—
IRB-11	MINO	Domain/bound	workout	insurancerecovery	insurancerecovery >= 0 (logical bound: cannot be negative)	—
IRB-12	MINO	Domain/bound	workout	exposureatdefault	exposureatdefault >= 0 (logical bound: cannot be negative)	—
IRB-13	MINO	Domain/bound	workout	drawnbalance	drawnbalance >= 0 (logical bound: cannot be negative)	—
IRB-14	MINO	Domain/bound	workout	timetoresolution	timetoresolution >= 0 (logical bound: cannot be negative)	—
IRB-18	MINO	Domain/bound	facility	termmonths	termmonths > 0 (positive tenor)	—
IRB-19	MINO	Domain/bound	facility	originalbalance	originalbalance >= 0 (logical bound)	—
IRB-20	MINO	Domain/bound	facility	originalpropertyvalue	originalpropertyvalue >= 0 (logical bound)	—
IRB-21	MINO	Domain/bound	facility	interestrate	interestrate >= 0 (logical bound)	—
IRB-22	MINO	Domain/bound	facility	ftbindicator	ftbindicator in {0,1}	—
IRB-23	MINO	Domain/bound	default	materialityindicator	materialityindicator in {0,1}	—
IRB-24	MINO	Conditional	default	dodbasis, dodregime	IF dodbasis = utp THEN dodregime present (basis needs a regime stamp)	—
IRB-26	MINO	Domain/bound	borrower	incomeatorigination	incomeatorigination >= 0 (logical bound)	—
IRB-28	MINO	Domain/bound	valuation	indexedflag	indexedflag in {0,1}	—

IFRS9 (11 rules)

ID	Sev	Type	Entity	Roles (semantic)	Rule	Reg reference
IFRS9-01	CRIT	Domain/bound	snapshot	<i>ifrs9stage</i>	<i>ifrs9stage</i> within its declared staging domain	—
IFRS9-02	CRIT	Conditional	snapshot	<i>dayspastdue</i> , <i>ifrs9stage</i>	IF <i>dayspastdue</i> > 90 THEN <i>ifrs9stage</i> = 3 (90+ DPD backstop -> Stage 3)	—
IFRS9-03	MATE	Conditional	snapshot	<i>dayspastdue</i> , <i>ifrs9stage</i>	IF <i>dayspastdue</i> > 30 THEN <i>ifrs9stage</i> >= 2 (30+ DPD SICR presumption)	—
IFRS9-05	MATE	Domain/bound	snapshot	<i>dayspastdue</i>	<i>dayspastdue</i> >= 0 (cannot be negative)	—
IFRS9-20	MATE	Conditional	snapshot	<i>forbearanceindicator</i> , <i>ifrs9stage</i>	IF <i>forbearance</i> THEN <i>ifrs9_stage</i> >= 2 (<i>forbearance</i> is a SICR trigger)	IFRS 9 B5.5.17; EBA/GL/2018/06
IFRS9-21	MATE	Conditional	snapshot	<i>ifrs9stage</i> , <i>dayspastdue</i>	IF <i>ifrs9stage</i> = 1 THEN <i>dayspastdue</i> <= 30 (Stage 1 excludes 30+ DPD)	IFRS 9 5.5.11 (30 DPD presumption)
IFRS9-04	MINO	Conditional	snapshot	<i>ifrs9stage</i> , <i>dayspastdue</i>	IF <i>ifrs9stage</i> in {2,3} THEN <i>dayspastdue</i> present (arrears measured)	—
IFRS9-08	MINO	Domain/bound	snapshot	<i>forbearanceindicator</i>	<i>forbearanceindicator</i> in {0,1}	—
IFRS9-09	MINO	Domain/bound	snapshot	<i>restructureindicator</i>	<i>restructureindicator</i> in {0,1}	—
IFRS9-22	MINO	Conditional	snapshot	<i>ifrs9stage</i> , <i>dayspastdue</i>	IF <i>ifrs9stage</i> = 3 THEN <i>dayspastdue</i> present (credit-impaired measured)	IFRS 9 Appendix A (credit-impaired)
IFRS9-23	MINO	Conditional	snapshot	<i>forbearanceindicator</i> , <i>restructureindicator</i> , <i>dayspastdue</i>	IF <i>forbearance</i> THEN <i>restructure</i> flag or arrears (not a clean Stage 1)	EBA/GL/2018/06 (<i>forbearance</i> classification)

IRB+IFRS9 (2 rules)

ID	Sev	Type	Entity	Roles (semantic)	Rule	Reg reference
IFRS9-06	MATE	Domain/bound	snapshot	<i>currentbalance</i>	<i>currentbalance</i> >= 0 (logical bound)	—
IFRS9-07	MATE	Identity/derivation	snapshot	<i>indexedltv</i> , <i>currentbalance</i> , <i>indexedpropertyvalue</i>	<i>indexedltv</i> == <i>currentbalance</i> / <i>indexedpropertyvalue</i> * 100 (+/- 0.5)	—

Encoded exceptions & notes

- **IFRS9-01** — Domain defaults to {1,2,3}; override per dataset via mapping key '*stagedomain*' (e.g. [0,1,2,3] for a 0-based DPD bucket).
- **IFRS9-02** — Rebuttable 90+ DPD backstop; encode institutional rebuttals as exceptions.
- **IFRS9-03** — Rebuttable 30+ DPD SICR presumption; encode rebuttals as exceptions.
- **IFRS9-20** — Rebuttable where *forbearance* did not increase credit risk; encode as exception.
- **IFRS9-23** — Soft cross-check; *forborne* with 0 DPD and no *restructure* flag is unusual.

- **IRB-03** — 'open' workouts are censored and out of scope by construction.
- **IRB-04** — Cured workouts close without a property sale; no sale_date expected.
- **IRB-33** — Factor 1.5 allows fees/arrears capitalisation; tune per portfolio.
- **IRB-35** — Wide LOGICAL bound, not typical [0,1]; typical-range checks route to distributional test.

Verdict logic (all rules)

- **Scope** = rows meeting the IF-condition, minus rows whose participating values are censored/N-A (value-semantics).
- **Exceptions** are removed from violations and reported with counts (silent exceptions are how rules rot).
- **Gate:** empty scope → NOT-APPLICABLE; rate $\leq$ tolerance → PASS; rate > tolerance → VIOLATION.
- **Pattern read** on each violation: lift-based CLUSTERED vs SCATTERED (clustered ⇒ feed/segment fault; scattered ⇒ capture error).
- Severity orders the report and labels the finding; it never changes pass/fail math.